

JEE MAIN 2020 3rd SEPT SHIFT 2

Question 1. Dimension of solar constant is:

(1) $M^1 L^0 T^{-3}$

(2) $M^1 L^1 T^{-3}$

(3) $M^0 L^0 T^3$

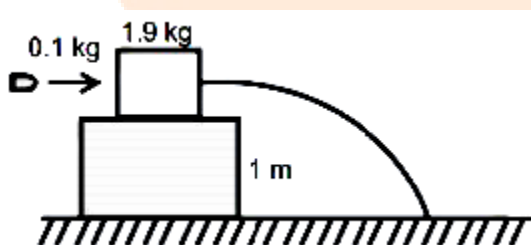
(4) $M^1 L^2 T^{-3}$

Ans. (1)

$$\text{Sol. Solar constant} = \frac{\text{Energy}}{\text{Time Area}}$$

$$= \frac{M^1 L^2 T^{-2}}{T L^2} = M^1 L^0 T^{-3}$$

Question 2. In the given diagram a 0.1 kg bullet moving with speed 20 m/sec strikes 1.9 kg mass and get embedded in it. Find the kinetic energy of the mass with which it will strikes the ground is.



(1) 11 J

(2) 21 J

(3) 25 J

(4) 30 J

Ans. (2)

Sol. Conservation of linear momentum

$$0.1 \times 20 = (0.1 + 1.9) \times v$$

$$v = 1 \text{ m/s}$$

Using work energy theorem

$$W_g = \Delta k$$

$$2 \times g \times 1 = k - \frac{1}{2} \times 2 \times 1^2$$

$$\therefore k = 21 \text{ J}$$

Question 3. A mass m is moving in SHM on a line with amplitude A and frequency f in a spring-mass system. Suddenly half of the mass comes to rest just at the moment, when it crosses mean position, then the new amplitude becomes λA , then λ will be

(1) $\frac{1}{2}$

(2) $\frac{1}{\sqrt{2}}$

(3) $\sqrt{2}$

(4) 1

Ans. (3)

Sol. As the external force is absent, by conservation of linear momentum

$$m_i u_i = m_f v_f$$

$$m.A\omega = \frac{m}{2} A' \omega'$$

$$m.A\sqrt{\frac{k}{m}} = \frac{m}{2} A' \sqrt{\frac{k}{m/2}}$$

$$A = \frac{1}{2} \sqrt{2} A'$$

$$A' = \sqrt{2} A$$

Question 4. A loop of area 'S' m² and N turns carrying current 'i' is placed in a uniform magnetic field 'B' with its plane parallel to $\vec{B}$. If torque ' τ ' is experienced by loop due to magnetic field, find $|\vec{B}|$

(1) $\frac{\tau}{NiS}$

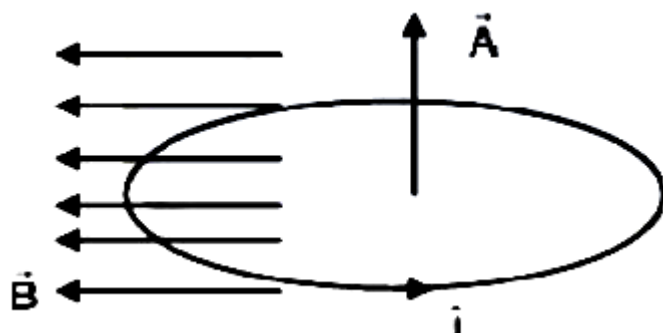
(2) $\frac{N\tau}{iS}$

(3) $\frac{i\tau}{NS}$

(4) $\frac{S\tau}{Ni}$

Ans. (1)

Sol.

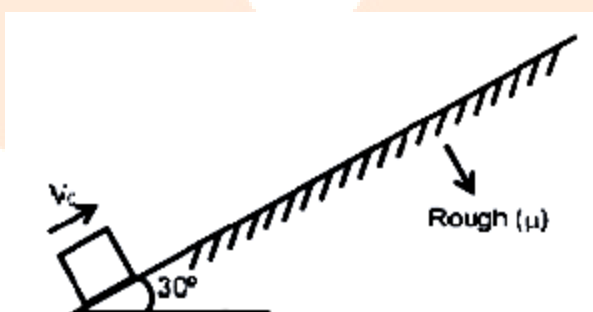


$$\tau = |\vec{M} \times \vec{B}| = NiAB \sin(90^\circ)$$

$$= NiAB = NiSB$$

$$\Rightarrow B = \frac{\tau}{NiS}$$

Question 5. A block starts going up a rough inclined plane with speed V_0 as shown in figure. After some time reaches to starting point again, with a speed $\frac{V_0}{2}$. Find coefficient of friction ' μ '. Given $g = 10 \text{ m/s}^2$.



(1) 0.15

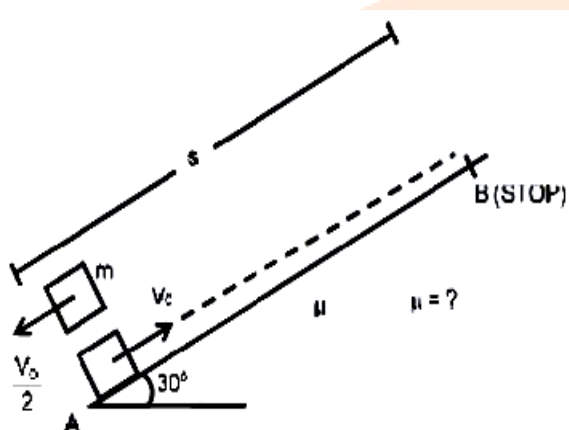
(2) 0.35

(3) 0.75

(4) 0.80

Ans. (2)

Sol.



A to B

$$a_1 = g \sin 30^\circ + \mu g \cos 30^\circ$$

$$= \frac{g}{2} + \frac{\mu g \sqrt{3}}{2}; g = 10 \text{ m/s}^2$$

$$V_0^2 - 2a_1(s) = 0$$

$$s = \frac{V_0^2}{2a_1} \dots (i)$$

B to A

$$a_2 = \frac{g}{2} - \frac{\mu \sqrt{3}}{2} g$$

$$\left(\frac{V_0}{2} \right)^2 = 2a_2(s)$$

$$s = \frac{V_0^2}{8a_2} \quad \dots \text{ (ii)}$$

From equation (i) and (ii)

$$\frac{V_0^2}{a_1} = \frac{V_0^2}{4a_2}$$

$$\Rightarrow a_1 = 4a_2$$

$$\Rightarrow 5 + 5\sqrt{3}\mu = 4(5 - 5\sqrt{3}\mu)$$

$$\Rightarrow 5 + 5\sqrt{3}\mu = 20 - 20\sqrt{3}\mu \Rightarrow 25\sqrt{3}\mu = 15$$

$$\Rightarrow \mu = \frac{\sqrt{3}}{5} = 0.35$$

Question 6. An ideal gas is heated by 160 J at constant pressure, its temperature rises by 50°C and if 240 J of heat is supplied at constant volume, temperature rises by 100°C , then its degree of freedom should be:

(1) 3

(2) 5

(3) 6

(4) 7

Ans. (3)

Sol. At constant pressure:

$$\Delta Q = nC_p \Delta T$$

$$\Rightarrow 160 = nC_p \cdot 50 \quad \dots (1)$$

At constant volume

$$\Delta Q = nC_v \Delta T$$

$$\Rightarrow 240 = nC_v \cdot 100 \quad \dots (2)$$

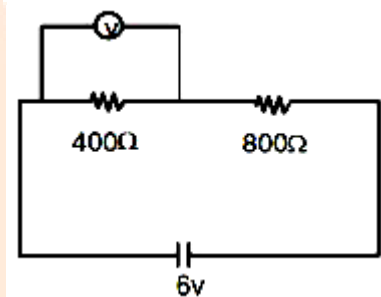
Equation (1) divided by (2)

$$\frac{160}{240} = \frac{C_p}{C_v} \cdot \frac{50}{100}$$

$$\frac{C_p}{C_v} = \frac{4}{3} = 1 + \frac{2}{f}$$

$$f = 6$$

Question 7. In the given diagram resistance of voltmeter is $10 \text{ k}\Omega$. Find reading of the voltmeter.



(1) 4v

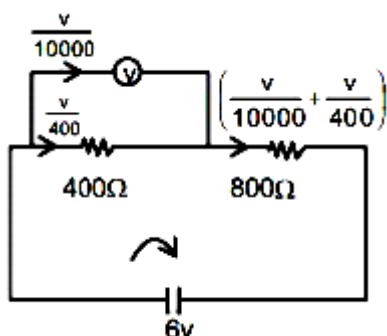
(2) 3.23v

(3) 1.95v

(4) 1.26v

Ans. (3)

Sol.



Let voltmeter reading is v

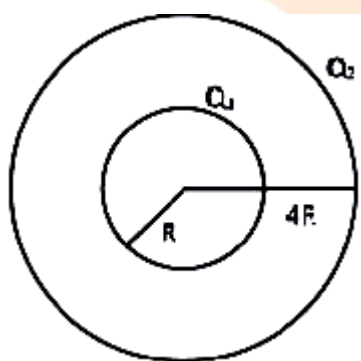
$$\frac{v}{400} \times 400 + \left(\frac{v}{10000} + \frac{v}{400} \right) 800 = 6$$

$$\Rightarrow v + \frac{8v}{100} + 2v - 6$$

$$\frac{77v}{25} = 6$$

$$v = \frac{150}{77} = 1.95v$$

Question 8. In the given figure, there are two concentric spherical shells, find potential difference between the spheres



$$(1) \frac{3}{8\pi \epsilon_0} \cdot \frac{Q_1}{R}$$

$$(2) \frac{3}{16\pi \epsilon_0} \cdot \frac{Q_2}{R}$$

$$(3) \frac{3}{4\pi \epsilon_0} \cdot \frac{Q_1}{R}$$

$$(4) \frac{3}{16\pi \epsilon_0} \cdot \frac{Q_1}{R}$$

Ans. (4)

$$\text{Sol. } V_{\text{inner}} = \frac{KQ_1}{R} + \frac{KQ_2}{4R}$$

$$V_{\text{outer}} = \frac{KQ_1}{4R} + \frac{KQ_2}{4R}$$

Potential difference

$$\begin{aligned} \Delta V &= V_{\text{inner}} - V_{\text{outer}} \\ &= \frac{3}{4} \cdot \frac{KQ_1}{R} = \frac{3}{16\pi \epsilon_0} \cdot \frac{Q_1}{R} \end{aligned}$$

Question 9. A body cools from 50°C to 40°C in 5 minutes in surrounding temperature 20°C . Find temperature of body in next 5 minutes.

$$(1) 13.3^\circ\text{C}$$

$$(2) 23.3^\circ\text{C}$$

(3) 43.3°C

(4) 33.3°C

Ans. (4)

Sol. Using Newton's Law of cooling

$$\frac{50 - 40}{5 \text{ Min}} = K \left(\frac{50 + 40}{2} - 20 \right) \quad \dots\dots (i)$$

Next 5 Min.

$$\frac{40 - \theta}{5} = K \left(\frac{40 + \theta}{2} - 20 \right) \quad \dots\dots (ii)$$

Dividing (ii) / (i)

$$\frac{40 - \theta}{10} = \frac{40 + \theta - 40}{50 + 40 - 40} = \frac{\theta}{50}$$

$$40 - \theta = \frac{\theta}{5}$$

$$200 - 5\theta = \theta$$

$$\therefore \theta = \frac{200}{6} = 33.3^\circ\text{C}$$

Question 10. A square wire loop of side 30 cm & wire cross section having diameter 4 mm is placed perpendicular to a magnetic field changing at the rate 0.2 T/s. Find induced current in the wire loop.

(Given: Resistivity of wire material is $1.23 \times 10^{-8} \Omega\text{m}$)

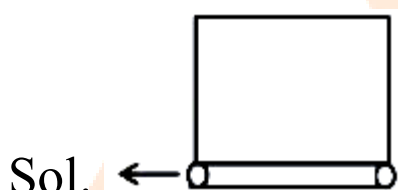
(1) $5.34 \times 10^2 \text{ A}$

(2) $3.34 \times 10^2 \text{ A}$

(3) $7.34 \times 10^2 \text{ A}$

(4) $1.34 \times 10^2 \text{ A}$

Ans. (1)



$$R_{\text{wire}} = \frac{\rho l}{A}$$

$$\phi = BA; |e| = \frac{d\phi}{dt} = \frac{dB}{dt} (A)$$

$$i = \frac{e}{R} = \left| \frac{dB}{dt} \right| \frac{(A)^2}{\rho \times L} = \frac{0.2 \times \{\pi \times 2 \times 10^{-3}\}^2}{1.23 \times 10^{-8} \times 4 \times 0.3}$$

$$= \frac{0.2 \times \pi^2 \times 10^{-6}}{1.23 \times 10^{-8} \times 0.3}$$

$$= 5.34 \times 10^2 \text{ A}$$

Question 11. Electric field of an electromagnetic wave is $\vec{E} = E_0 \cos(\omega t - kx) \hat{j}$. The equation of corresponding magnetic field at $t = 0$ should be:

(1) $\vec{B} = E_0 \sqrt{\mu_p \epsilon_0} \cos k(\hat{k})$

$$(2) \vec{B} = \frac{E_0}{\sqrt{\mu_0 \epsilon_0}} \cos k \times \hat{k}$$

$$(3) \vec{B} = E_0 \sqrt{\mu_0 \epsilon_0} \cos kx (-\hat{k})$$

$$(4) \vec{B} = \frac{E_0}{\sqrt{\mu_0 \epsilon_0}} \cos k \times (-\hat{k})$$

Ans. (1)

$$\text{Sol. } B_0 = \frac{E_0}{C} = \frac{E_0}{1 / \sqrt{\mu_0 \epsilon_0}} = E_0 \sqrt{\mu_0 \epsilon_0}$$

As the light is propagating in x direction

$$\& \hat{E} \times \hat{B} \parallel \hat{C}$$

$\therefore \hat{B}$ should be in $\hat{k}$ direction

Hence, option (1) is correct.

Question 12. Given two points sources having same power of 200W. One source is emitting photons of $\lambda_1 = 500 \text{ nm}$ and other emitting X-ray photons of $\lambda_2 = 1 \text{ nm}$. Find ratio of photon density from both the sources?

(1) 200

(2) 500

(3) 250

(4) 0.4

Ans. (2)

Sol. P_s - Power of source

$$P_s = n \frac{hc}{\lambda}; n = \text{no. of photons emitted/s}$$

$$\Rightarrow n \propto \lambda \Rightarrow \frac{n_1}{n_2} = \frac{\lambda_1}{\lambda_2} = 500$$

Question 13. Image of a point object placed on principal axis at a distance of 30 cm from the mirror is formed at 10 cm from the mirror. If object start moving with velocity 9 cm/sec. Find initial velocity of image.

(1) -9 cm

(2) -4 cm

(3) -1 cm

(4) -3 cm

Ans. (1)

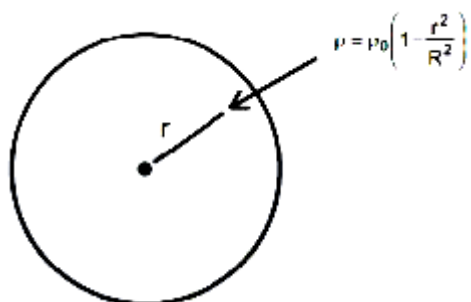
Sol. Velocity of image

$$v_i = -\frac{v^2}{u^2} \times v_o$$

$$= -\frac{10^2}{30^2} (9)$$

$$= -1 \text{ cm/sec}$$

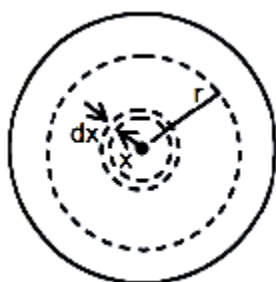
Question 14. Mass density of a sphere having radius R varies as $\rho = \rho_0 \left(1 - \frac{r^2}{R^2}\right)$. Find maximum magnitude of gravitational field.



- (1) $\frac{4}{3} \pi G \rho_0 R$
- (2) $\frac{2\sqrt{3}}{5} \pi G \rho_0 R$
- (3) $\frac{8\sqrt{5}}{27} \pi G \rho_0 R$
- (4) $\frac{2\sqrt{5}}{27} \pi G \rho_0 R$

Ans. (3)

Sol.



$$dm = \rho \times 4\pi x^2 dx$$

$$= \rho_0 \left(1 - \frac{x^2}{R^2} \right) \times 4\pi x^2 dx$$

$$m = 4\pi\rho_0 \int_0^r \left(x^2 - \frac{x^4}{R^2} \right) dx$$

$$m = 4\pi\rho_0 \left[\frac{r^3}{3} - \frac{r^5}{5R^2} \right]$$

$$E = \frac{Gm}{r^2}$$

$$= \frac{G}{r^2} \times 4\pi\rho_0 \left(\frac{r^3}{3} - \frac{r^5}{5R^2} \right)$$

$$E = 4\pi G\rho_0 \left(\frac{r}{3} - \frac{r^3}{5R^2} \right)$$

E is maximum when

$$\frac{dE}{dr} = 0 \Rightarrow \frac{dE}{dr} = 4\pi G\rho_0 \left(\frac{1}{3} - \frac{3r^2}{5R^2} \right) = 0$$

$$\Rightarrow r = \frac{\sqrt{5}}{3} R$$

$$E_{\max} = 4\pi G\rho_0 \times \frac{\sqrt{5}R}{3} \left[\frac{1}{3} - \frac{1}{5} \times \frac{5}{9} \right]$$

$$E_{\max} = \frac{8\sqrt{5}}{27} \pi G\rho_0 R$$

Question 15. Two light rays having the same wavelength λ in vacuum are in phase initially. Then the first ray travels a path L_1 through a medium of refractive index n_1 while the second ray travels a path of length L_2 through a medium of refractive index n_2 . The two waves are then combined to produce interference. The phase difference between the two waves at the point of interference is:

(1) $\frac{2\pi}{\lambda}(L_2 - L_1)$

(2) $\frac{2\pi}{\lambda}(n_1 L_1 - n_2 L_2)$

(3) $\frac{2\pi}{\lambda}(n_2 L_1 - n_1 L_2)$

(4) $\frac{2\pi}{\lambda}\left(\frac{L_1}{n_1} - \frac{L_2}{n_2}\right)$

Ans. (2)

Sol. The optical path between any two points is proportional to the time of travel. The distance traveled by light in a medium of refractive index μ in time t is given by

$$d = vt \quad \dots (i)$$

where v is velocity of light in the medium. The distance traversed by light in a vacuum in this time, $\Delta = ct$

$$= c \times \frac{d}{v} \quad [\text{from equation (i)}]$$

$$= d \frac{c}{v} = \mu d \quad \dots\dots (ii) \quad \left(\text{Since, } \mu = \frac{c}{v} \right)$$

This distance is the equivalent distance in vacuum and is called optical path.

Here, optical path for first ray = $n_1 L_1$

Optical path for second ray = $n_2 L_2$

Path difference = $n_1 L_1 - n_2 L_2$

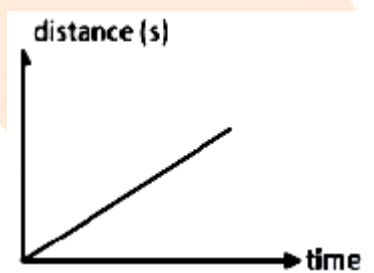
Now phase difference

$$= \frac{2\pi}{\lambda} \times \text{path difference}$$

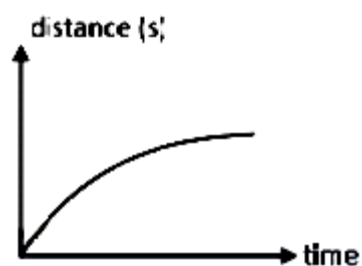
$$= \frac{2\pi}{\lambda} \times (n_1 L_1 - n_2 L_2)$$

Question 16. Constant power P is supplied to particle having mass m , initially at rest. Choose correct graph.

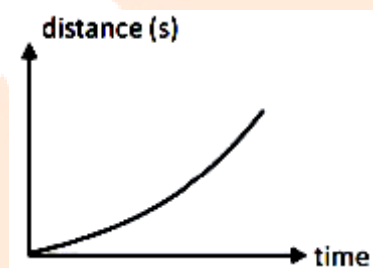
(1)



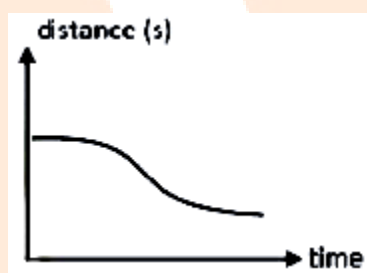
(2)



(3)



(4)



Ans. (3)

$$\text{Sol. } P \cdot t = \frac{1}{2} m v^2 \Rightarrow v = \left(\sqrt{\frac{2P}{m}} \right) t^{1/2}$$

$$s = \int_0^t v dt = \sqrt{\frac{2P}{m}} \int_0^t t^{1/2} dt$$

$$= \sqrt{\frac{2P}{m}} \cdot \frac{t^{3/2}}{3/2}$$

$$s = \sqrt{\frac{8P}{9m}} \cdot t^{3/2}$$

Question 17. A p - n junction becomes active as photons of wavelength; $\lambda = 400 \text{ nm}$ falls on it. Find the energy band gap?

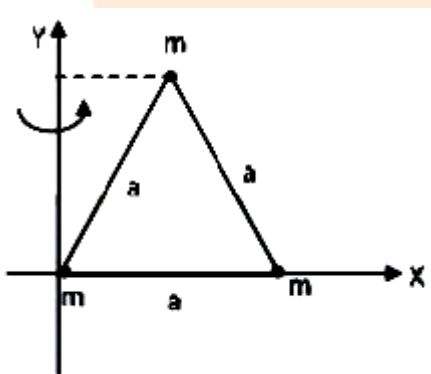
- (1) 3.09 eV
- (2) 4.51 eV
- (3) 2.45 eV
- (4) 5.34 eV

Ans. (1)

Sol. $\lambda = 400 \text{ nm}$

$$\text{Band gap } E_g = \frac{hc}{\lambda} = \frac{1237.5}{400} = 3.09 \text{ eV}$$

Question 18. In the diagram three point masses 'm' each are fixed at the corners of an equilateral triangle. Moment of inertia of the system about y-axis is $\frac{N}{20} ma^2$, N is:



- (1) 25
- (2) 50

(3) 75

(4) 100

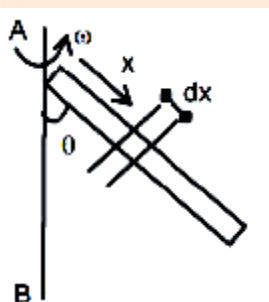
Ans. (1)

$$\text{Sol. } I = m \times O^2 + ma^2 + m\left(\frac{a}{2}\right)^2$$

$$= \frac{5}{4}ma^2 = \frac{25}{20}ma^2$$

$$N = 25$$

Question 19. A rod is rotating with angular velocity ω about axis AB. Find $\cos \theta$



(1) $\frac{g}{2\ell\omega^2}$

(2) $\frac{g}{\ell\omega^2}$

(3) $\frac{2g}{\ell\omega^2}$

(4) $\frac{3g}{2\ell\omega^2}$

Ans. (4)

Sol. Torque of centrifugal force

$$\tau_{cf} = dm \cdot x \sin \theta \omega^2 x \cos \theta = \frac{m}{\ell} \omega^2 \sin \theta \cos \theta \int_0^\ell x^2 dx$$

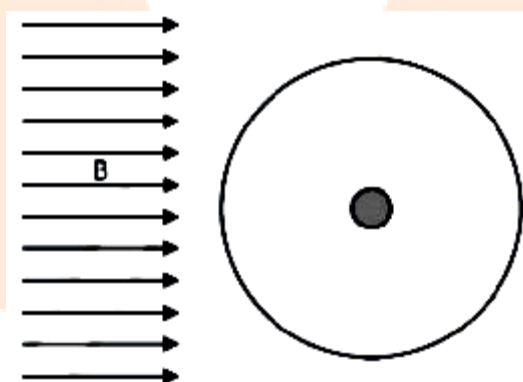
$$\tau_{cf} = \frac{m \ell^2 \omega^2 \sin \theta \cos \theta}{3}$$

$$\tau_{mg} = \tau_{cf}$$

$$mg \frac{\ell}{2} \sin \theta = \frac{m \ell^2 \omega^2 \sin \theta \cos \theta}{3}$$

$$\cos \theta = \frac{3g}{2\ell \omega^2}$$

Question 20. In a diamagnetic sphere a cavity is made at its centre and now paramagnetic material is inserted in the cavity. The sphere is kept in a external magnetic field at centre.



(1) 0

(2) B

$$(3) B_0 > B$$

$$(4) B_0 < B$$

Ans. (1)

Sol. When magnetic field is applied diamagnetic substance produces magnetic field in opposite direction so net magnetic field will be zero.

Chemistry

Question 21. A mixture of one mole of each of $O_2(g)$, $H_2(g)$, $He(g)$ exists in a container of volume V at temperature T , in which partial pressure of $H_2(g)$ is 2 atm. The total pressure in the container is:

$$(1) 6 \text{ atm}$$

$$(2) 18 \text{ atm}$$

$$(3) 33 \text{ atm}$$

$$(4) 24 \text{ atm}$$

Answer: (1)

Solution:

$$P_{\text{gas}} = \frac{n_{\text{gas}} RT}{V}$$

as n , t & V constant So

$$P_{H_2} = P_{O_2} = P_{He} = 2 \text{ atm}$$

$$\text{So, } P_{\text{Total}} = P_{\text{H}_2} + P_{\text{O}_2} + P_{\text{He}}$$

$$= 6\text{atm}$$

Question 22. What is the concentration and %(w/w) of 5.6 V H_2O_2 solution? [Given molar mass of $\text{H}_2\text{O}_2 = 34 \text{ g/mol}$ density = 1 g/mL]

(1) 0.5, 1.70

(2) 0.25, 1.70

(3) 0.5, 0.85

(4) 0.25, 0.85

Answer: (1)

Solution:

For H_2O_2

$$\text{Molarity} = \frac{\text{Volume strength}}{11.2} = \frac{5.6}{11.2} = 0.5 \text{ M}$$

$$\text{Molarity} = \frac{\%(\text{w/w}) \times 10 \times d}{\text{GMM}}$$

$$0.5 = \frac{\%(\text{w/w}) \times 10 \times d}{34}$$

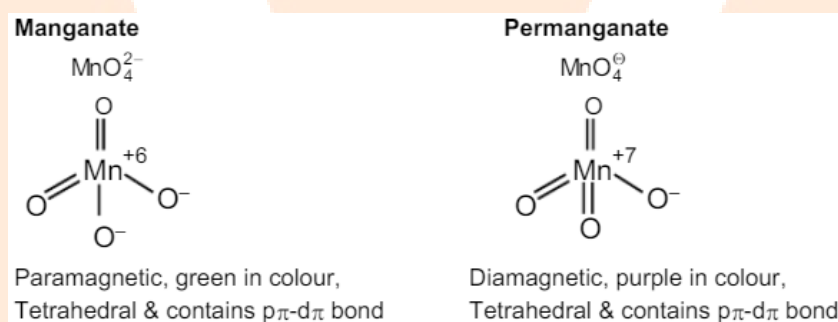
$$\%(\text{w/w}) = \frac{0.5 \times 34}{10} = 1.7$$

Question 23. Find incorrect statement about manganate and permanganate ions.

- (1) Both manganate and permanganate ions are Paramagnetic
- (2) Manganate ion is green in colour while permanganate ion is purple in colour
- (3) Both manganate and permanganate ions have tetrahedral shape
- (4) In manganate and permanganate ions Mn forms $p\pi$ - $d\pi$ bond with oxygen.

Answer: (1)

Solution:



Question 24. Among the following statements identify the correct set of statements

- (a) Size of Be is smaller than Mg
- (b) Ionisation energy of Be is greater than Al
- (c) Both Be and Al form covalent compounds readily
- (d) Both Be and Al does not react with nitrogen

(1) a, b, c

(2) a, c, d

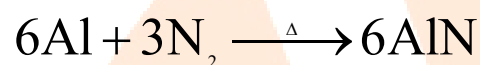
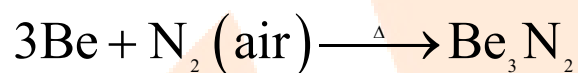
(3) b, c, d

(4) a, b, d

Answer: (1)

Solution:

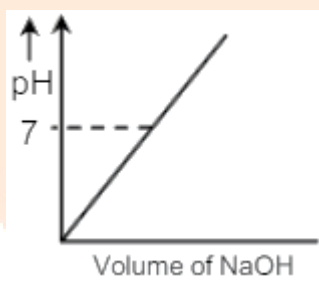
Both Be and Al react with nitrogen to form nitride



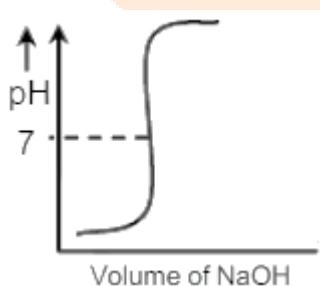
Remaining all statements are correct.

Question 25. In 0.1 M HCl solution, 0.1 M NaOH solution is added gradually then identify the correct graph this titration.

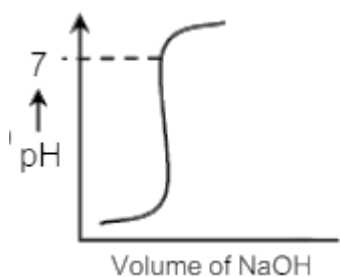
(1)



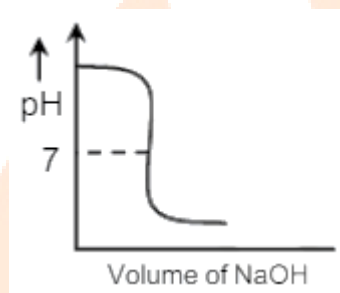
(2)



(3)



(4)



Answer: (2)

Solution:

At equivalence point pH is 7 and pH increases with addition of NaOH so correct graph is (b).

Question 26. For the reaction, $2A + 3B + \frac{3}{2}C \rightarrow 3P$

The correct relation between the rate of reaction of species A, B and C is:

$$(1) \frac{dn_A}{dt} = \frac{2}{3} \frac{dn_B}{dt} = \frac{4}{3} \frac{dn_C}{dt}$$

$$(2) 2 \frac{dn_A}{dt} = 3 \frac{dn_B}{dt} = \frac{3}{2} \frac{dn_C}{dt}$$

$$(3) \frac{3}{2} \frac{dn_A}{dt} = \frac{dn_B}{dt} = \frac{3}{4} \frac{dn_C}{dt}$$

$$(4) \frac{dn_A}{dt} = \frac{dn_B}{dt} = \frac{dn_C}{dt}$$

Answer: (1)

Solution:

For a given reaction

$$\text{rate} = -\frac{1}{2} \frac{dn_A}{dt} = -\frac{1}{3} \frac{dn_B}{dt} = -\frac{2}{3} \frac{dn_C}{dt}$$

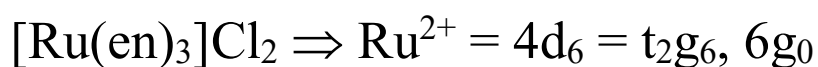
$$\text{rate} = \frac{dn_A}{dt} = \frac{2}{3} \frac{dn_B}{dt} = \frac{4}{3} \frac{dn_C}{dt}$$

Question 27. The crystal field configuration of complexes $[\text{Ru}(\text{en})_3]\text{Cl}_2$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ respectively is:

- (1) t_{2g}^4, e_g^2 and t_{2g}^6, e_g^0
- (2) t_{2g}^6, e_g^0 and t_{2g}^4, e_g^0
- (3) t_{2g}^4, e_g^2 and t_{2g}^4, e_g^2
- (4) t_{2g}^6, e_g^0 and t_{2g}^6, e_g^0

Answer: (2)

Solution:



So, correct answer is (2).

Question 28. What is the valency of an atom if its successive ionisation energies respectively are 800, 25356, 32456 kJ/mole?

(1) 3

(2) 4

(3) 5

(4) 6

Answer: (1)

Solution:

As difference in 3rd and 4th ionisation energies is high so atom contains 3 valence electrons. Hence valency of the atom is 3.

Question 29. For a hypothetical case let value of l is define as 0, 1, 2, 3($n + 1$) for principle quantum number n

(1) Atomic number of 1st noble gas is 8

(2) Atomic number of 1st alkali metal is 9

(3) Carbon has electron in 2pz

(4) Element with atomic number 13 has half-filled valence shell

Answer: (4)

Solution:

For $n = 1$ valence of $l = 0, 1, 2$ Electronic configuration = $1s^2 1p^6 1d^{10}$

(1) 1st noble gas atomic number is 18

(2) 1st alkali metal electronic configuration $\Rightarrow$
 $1s^2 1p^6 1d^{10} 2s^1 \Rightarrow (Z = 19)$

(3) Electronic configuration of C ($Z = 6$) $\Rightarrow 1s^2 1p^4$

(4) $Z = 13$, Electronic configuration = $1s^2 1p^6 1d^5$

So it has half-filled electronic configuration.

Question 30. A current of 2A is passed through a dichromate solution for 5 min. Then 0.104 g of Cr^{3+} ions are formed. What is the percentage efficiency of cell?

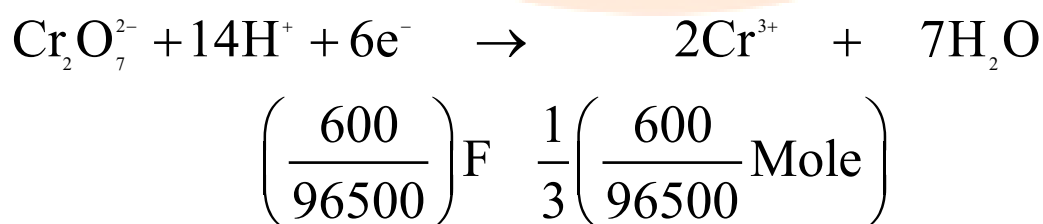
[Given $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$, Atomic mass of Cr = 52]

Answer: 96.50

Solution:

Charge (q) = $it = 2 \times 5 \times 60$

$$= 600\text{C} = \left(\frac{600}{96500} \text{F} \right)$$



$$\text{Theoretical mass of Cr}^{3+} = \frac{1}{3} \times \frac{600}{96500} \times 52g$$

So, efficiency =

$$\frac{W_{\text{actual}}}{W_{\text{Theoretical}}} \times 100 = \frac{0.104 \times 3 \times 96500}{52 \times 600} \times 100 = 96.50$$

Question 31. How much volume of 0.1 N NaOH will neutralize 10 mL of 0.1 N phosphoric acid?

Answer: 10.00

Solution:

Phosphoric acid is phosphorous acid (H_3PO_3).



For neutrization

$$(N_1 V_1)_{\text{acid}} = (N_2 V_2)_{\text{base}}$$

$$0.1 \times 10 = 0.1 \times (V_{\text{mL}})_{\text{NaOH}}$$

$$V_{\text{NaOH}} = 10 \text{ mL}$$

Question 32. In Isotonic solution of protein A and protein B, 0.73 gram of protein A is dissolved in 250 ml while 1.65 gram of protein B is dissolved in 1 L solution, then what is the ratio of molecular mass of Protein A and protein B?

Answer: 01.77

Solution:

For isotonic solution

$$I_1C_1 = i_2C_2 \quad \{\text{For protein } i = 1\}$$

$$C_1 = C_2$$

$$\Rightarrow \frac{0.73 \times 1000}{M_A \times 250} = \frac{1.65}{M_B \times 1}$$

$$\frac{M_A}{M_B} = \frac{0.73 \times 4}{1.65} = 1.77$$

Question 33. 6.022×10^{22} molecules of a compound X has mass 10 g. What is the molarity of solution containing 5g of 'X' in 2 Lit. solution, answer as P [Where $M = P \times 10^{-3}$ Mole/Lit]

Answer: 25.00

Solution:

$$\text{Number of mole of X} = \frac{6.022 \times 10^{22}}{6.022 \times 10^{23}} = \frac{10}{\text{Molar mass of X}}$$

So molar mass of X = 100 g

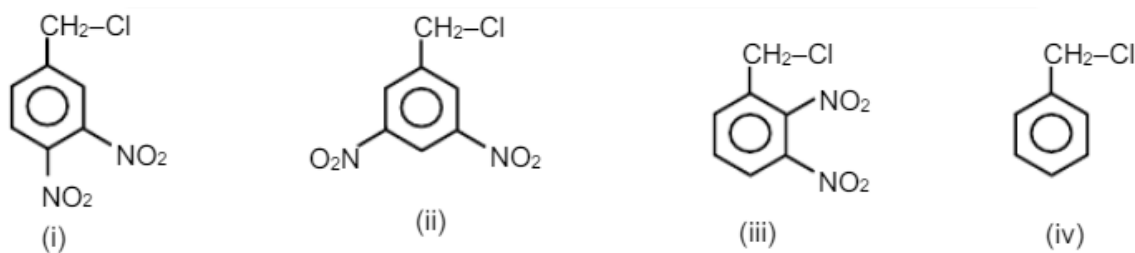
$$\text{Molarity} = \frac{5}{100 \times 2} = 0.025M$$

Ans. = 0.025 M

$$M = 25 \times 10^{-3}$$

So P = 25

Question 34. Write down nucleophilic substitution ($\text{S}_\text{N}2$) order for following.



(1) $\text{i} > \text{ii} > \text{iii} > \text{iv}$

(2) $\text{iv} > \text{iii} > \text{ii} > \text{i}$

(3) $\text{i} > \text{iii} > \text{ii} > \text{iv}$

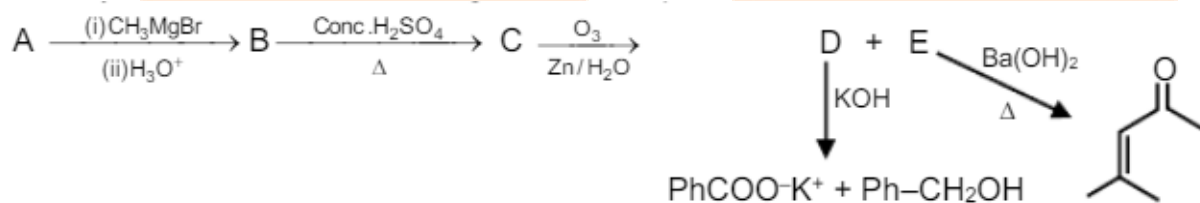
(4) $\text{iii} > \text{i} > \text{ii} > \text{iv}$

Answer: (3)

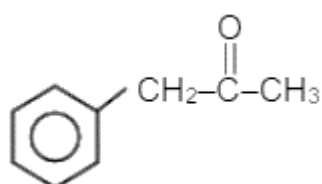
Solution:

$\text{S}_\text{N}2$ reaction depend upon – I, – M effect on substrate. On increase – I, – M, effect rate of $\text{S}_\text{N}2$ reaction increase.

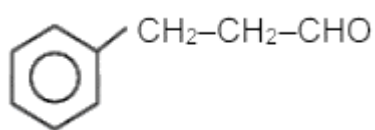
Question 35. Identify structure of A in following reaction sequence.



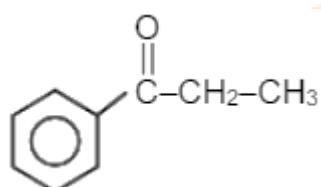
(1)



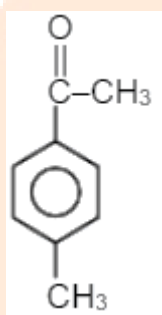
(2)



(3)

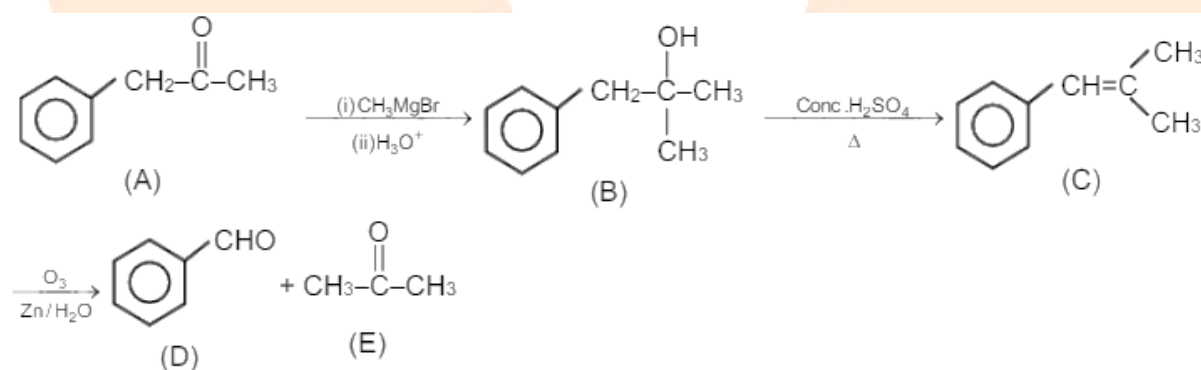


(4)



Answer: (1)

Solution:

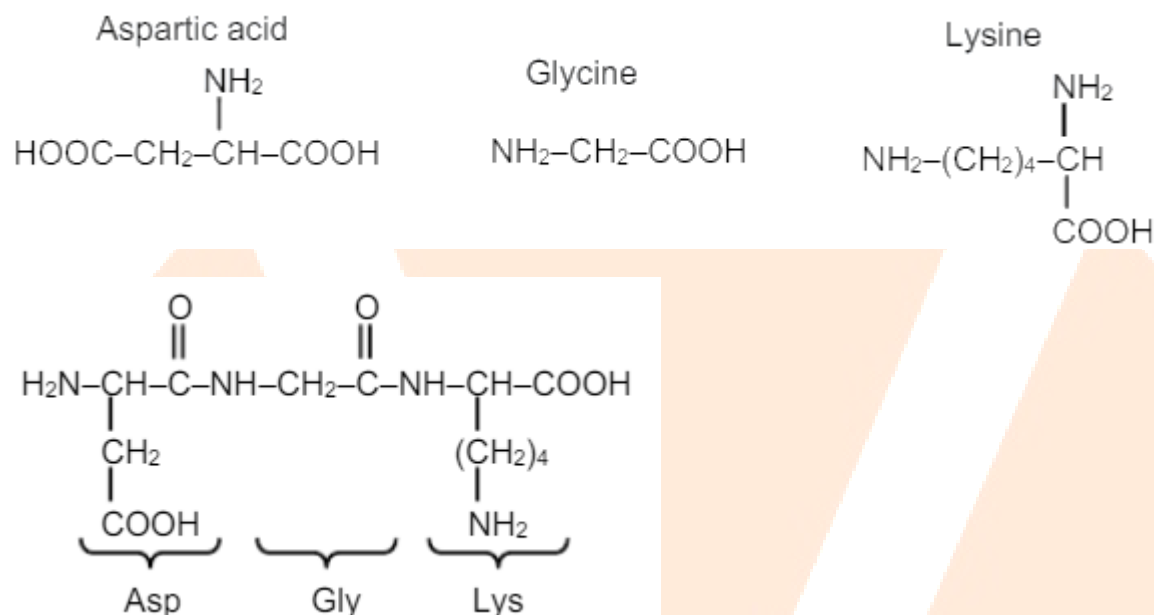


Question 36. Calculate $\text{—}\overset{\text{O}}{\parallel}{\text{C}}\text{—}$ in given structure of peptide chain.

Asp – Gly – Lys

Answer: 4.00

Solution:



Question 37. Match the columns

Column-I	Column-II
(A) Chloramphenicol	(a) Antacid
(B) Ranitidine	(2) Antihistamine
(C) Phenelzine (nardil)	(3) Antibiotic
(D) Morphine	(d) Analgesic
	(5) Antidepressant

(1) A-1; B-2; C-4; D-5

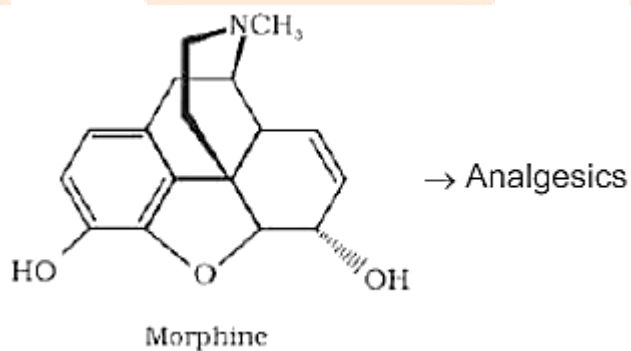
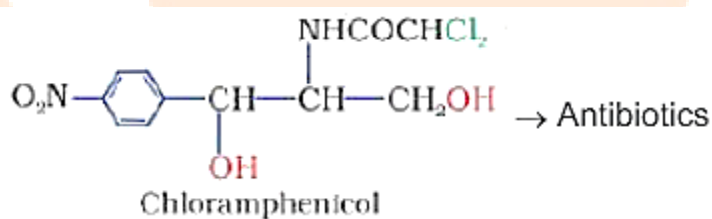
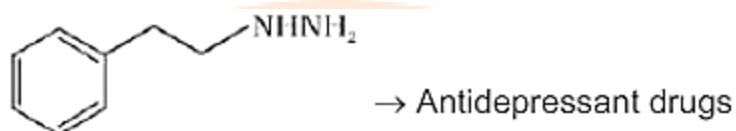
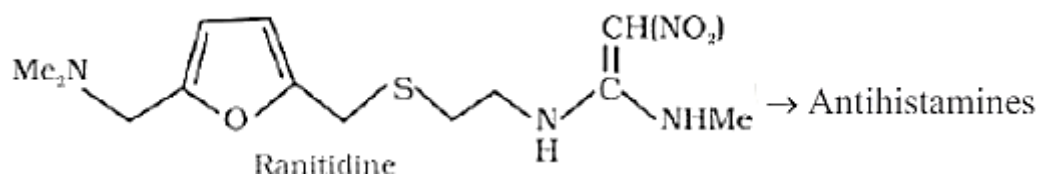
(2) A-3; B-2; C-5; D-4

(3) A-2; B-4; C-5; D-1

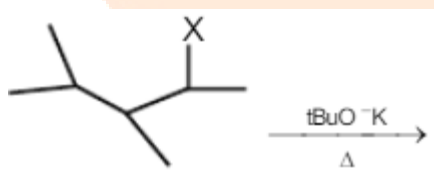
(4) A-3; B-2; C-1; D-5

Answer: (2)

Solution:

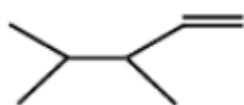


Question 38.

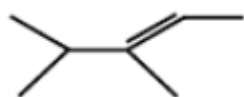


Find product of above reaction

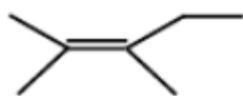
(1)



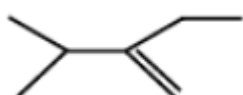
(2)



(3)

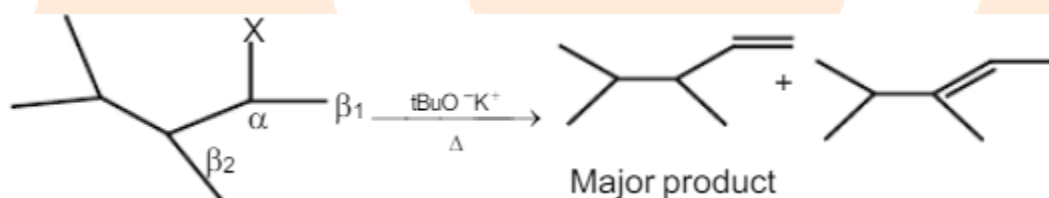


(4)



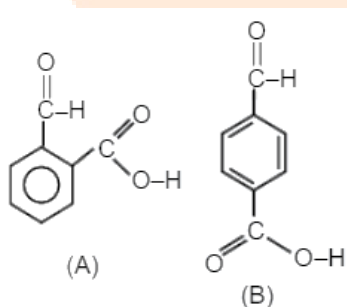
Answer: (1)

Solution:



This is an example of E_2 reaction and due to bulky base final product is Hoffmann alkene.

Question 39.



(A) B is less water soluble than A

(B) B is more crystalline in nature than A

(C) B has more boiling point than A

Select correct statement regarding above structures.

- (1) A, B are correct
- (2) B, C are correct
- (3) Only C are correct
- (4) A, B, C all are correct

Answer: (2)

Solution:

Due to inter molecular H-Bonding in B, than A, B is more soluble and having more B.P point than A.

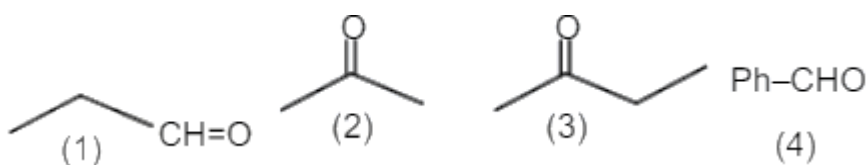
Question 40. Write down decreasing order of Nucleophilic addition reaction of following Propanal, Butanone, Propanone, Benzaldehyde

- (1) Propanal > Butanone > Propanone > Benzaldehyde
- (2) Propanal > Benzaldehyde > Propanone > Butanone
- (3) Propnone > Propanal > Butanone > Benzaldehyde
- (4) Propnone > Butanone > Benzaldehyde > Propanal

Answer: (2)

Solution:

Rate of NAR $\propto 1 - M$ on substate



$$1 > 4 > 2 > 3$$

Question 41. Which of the following statement are incorrect statement(s) for acid rain

- (A) It corrodes water pipes
- (B) It is not harmful for trees and plants
- (C) It does not cause breathing problem in human being and animals
- (D) It damages building and other structures made of stone or metal.

(1) A & B

(2) B & C

(3) A & C

(4) B & D

Answer: (2)

Solution:

(B) It is harmful for trees and plants

(C) It causes breathing problem in human being and animals.

Mathematics

Question 42. If $\int_0^{\frac{1}{2}} \frac{x^2}{(1-x^2)^{3/2}} dx = \frac{k}{6}$, then $k =$

(1) $3\sqrt{2} + \pi$

(2) $2\sqrt{3} - \pi$

(3) $2\sqrt{3} + \pi$

(4) $3\sqrt{2} - \pi$

Ans. (2)

Sol. $\frac{k}{6} = \int_0^{\frac{1}{2}} \frac{x^2}{(1-x^2)^{3/2}} dx$ $x = \sin \theta; dx = \cos \theta d\theta$

$$\Rightarrow \frac{k}{6} = \int_0^{\frac{\pi}{6}} \frac{\sin^2 \theta}{(1-\sin^2 \theta)^{3/2}} \cdot \cos \theta d\theta \Rightarrow \frac{k}{6} = \int_0^{\frac{\pi}{6}} \frac{\sin^2 \theta}{\cos^3 \theta} \cdot \cos \theta d\theta$$

$$\Rightarrow \frac{k}{6} = \int_0^{\frac{\pi}{6}} \tan^2 \theta d\theta = \int_0^{\frac{\pi}{6}} (\sec^2 \theta - 1) d\theta \Rightarrow$$

$$\frac{k}{6} = (\tan \theta - \theta) \Big|_0^{\pi/6} = \left(\frac{1}{\sqrt{3}} - \frac{\pi}{6} \right) = \frac{2\sqrt{3} - \pi}{6}$$

$$\Rightarrow k = 2\sqrt{3} - \pi.$$

Question 43. Let $\frac{x^2}{25} + \frac{y^2}{b^2} = 1$ and $\frac{x^2}{16} - \frac{y^2}{b^2} = 1$ are ellipse and hyperbola respectively such that $e_1 e_2 = 1$ where e_1 & e_2 are eccentricities. If distance between foci of ellipse is α and that of hyperbola is β then $(\alpha, \beta) =$

(1) (4, 5)

(2) (8, 10)

$$(3) (10, 7)$$

$$(4) (4, 10)$$

Ans. (2)

$$\text{Sol. } e_1 = \sqrt{1 - \frac{b^2}{25}}; e_2 = \sqrt{1 + \frac{b^2}{16}}$$

$$e_1 e_2 = 1$$

$$\Rightarrow (e_1 e_2)^2 = 1 \Rightarrow \left(1 - \frac{b^2}{25}\right) \left(1 + \frac{b^2}{16}\right) = 1$$

$$\Rightarrow 1 + \frac{b^2}{16} - \frac{b^2}{25} - \frac{b^4}{25 \times 16} = 1$$

$$\Rightarrow \frac{9}{16 \cdot 25} b^2 - \frac{b^4}{25 \cdot 16} = 0 \Rightarrow b^2 = 9$$

$$e_1 = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$$

$$e_2 = \sqrt{1 + \frac{9}{16}} = \frac{5}{4}$$

$$\alpha = 2(5)(e_1) = 8$$

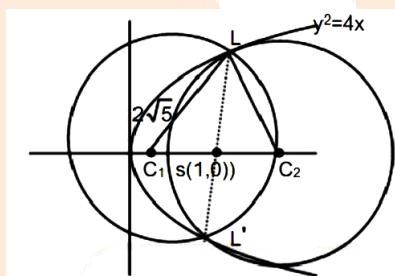
$$\beta = 2(4)(e_2) = 10$$

$$(\alpha, \beta) = (8, 10).$$

Question 44. Two equal circles of radius $2\sqrt{5}$ passes through the extremities of Latus Rectum of $y^2 = 4x$ then the distance between centres of circles is

- (1) 4
- (2) 8
- (3) 2
- (4) 6

Ans. (2)



Sol.

$$C_1C_2 = 2C_1S = 2\sqrt{20-4} = 8$$

Question 45. If

$$\int \sin^{-1} \left(\frac{\sqrt{x}}{\sqrt{1+x}} \right) dx = A(x) \tan^{-1} \sqrt{x} + B(x) + C, \text{ then } A(x) \text{ and}$$

$B(x)$ will be

- (1) $1+x, \sqrt{x}$
- (2) $1-x, -\sqrt{x}$
- (3) $1+x, -\sqrt{x}$
- (4) $1-x, \sqrt{x}$

Ans. (3)

$$\text{Sol. } I = \int \sin^{-1} \left(\frac{\sqrt{x}}{\sqrt{1+x}} \right) dx$$

$$\text{Let, } x = t^2 \Rightarrow dx = 2dt$$

$$I = \int 2t \tan^{-1} t dt$$

$$= 2 \left[\tan^{-1} t \cdot \frac{t^2}{2} - \int \frac{1}{1+t^2} \cdot \frac{t^2}{2} dt \right]$$

$$= 2 \left[\tan^{-1} t \cdot \frac{t^2}{2} - \frac{1}{2} \int \frac{1+t^2-1}{1+t^2} dt \right]$$

$$= t^2 \cdot \tan^{-1} t - \int \left(1 - \frac{1}{1+t^2} \right) dt$$

$$= t^2 \tan^{-1} t - t + \tan^{-1} t + C$$

$$= x \tan^{-1} \sqrt{x} - \sqrt{x} + \tan^{-1} \sqrt{x} + C$$

$$= (x+1) \tan^{-1} \sqrt{x} - \sqrt{x} + C \Rightarrow (Ax) = x+1 \text{ \& } B(x) = -\sqrt{x}$$

Question 46. The coefficient of term independent of x in the expansion of $\left(\frac{3x^2}{2} - \frac{1}{3x} \right)^9$ is λ , then the value of 18λ is

(1) 9

(2) 7

(3) 6

(4) 4

Ans. (2)

$$\begin{aligned}\text{Sol. } T_{r+1} &= {}^9C_r \left(\frac{3x^2}{2} \right)^{9-r} \left(-\frac{1}{3x} \right)^r \\ &= {}^9C_r \left(\frac{3}{2} \right)^{9-r} \left(-\frac{1}{3} \right)^r x^{18-3r}\end{aligned}$$

For the term independent of x put $18 - 3r = 0 \Rightarrow r = 6$

$$\Rightarrow T_7 = {}^9C_6 \left(\frac{3}{2} \right)^3 \left(-\frac{1}{3} \right)^6 = {}^9C_3 \left(\frac{1}{6} \right)^3 = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} \left(\frac{1}{6} \right)^3 = \left(\frac{7}{18} \right)$$

Question 47. If $|z_1 - 1| = \text{Re}(z_1)$, $|z_2 - 1| = \text{Re}(z_2)$ and $\arg(z_1 - z_2) = \frac{\pi}{3}$, then $\text{Im}(z_1 + z_2) =$

(1) $\frac{1}{\sqrt{3}}$

(2) $\frac{2}{\sqrt{3}}$

(3) $\frac{\sqrt{3}}{2}$

(4) $\sqrt{3}$

Ans. (2)

Sol. Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$

$$|z_1 - 1| = \operatorname{Re}(z_1)$$

$$(x_1 - 1)^2 + y_1^2 = x_1^2$$

$$y_1^2 - 2x_1 + 1 = 0 \quad \dots(1)$$

$$|z_2 - 1| = \operatorname{Re}(z_2)$$

$$(x_2 - 1)^2 + y_2^2 = x_2^2$$

$$y_2^2 - 2x_2 + 1 = 0 \quad \dots(2)$$

$$y_1^2 - y_2^2 - 2(x_1 - x_2) = 0$$

$$(y_1 - y_2)(y_1 + y_2) = 2(x_1 - x_2)$$

$$y_1 + y_2 = 2 \left(\frac{x_1 - x_2}{y_1 - y_2} \right) \quad \dots(3)$$

$$\text{Also, } \arg(z_1 - z_2) = \frac{\pi}{3}$$

$$\tan^{-1} \left(\frac{y_1 - y_2}{x_1 - x_2} \right) = \frac{\pi}{3}$$

$$\frac{y_1 - y_2}{x_1 - x_2} = \sqrt{3} \quad \dots(4)$$

$$\therefore y_1 + y_2 = \frac{2}{\sqrt{3}} \Rightarrow \operatorname{Im}(z_1 + z_2) = \frac{2}{\sqrt{3}}.$$

Question 48. The probability of 5 digit numbers that are made up of exactly two distinct digit is

(1) $\frac{135}{10^4}$

(2) $\frac{125}{10^4}$

(3) $\frac{144}{10^4}$

(4) $\frac{127}{10^4}$

Ans. (1)

Sol. Total = $9(10^4)$

Favourable = ${}^9C_2 (25 - 2) + {}^9C_1 (24 - 1) = 36(30) + 9(15) = 1080 + 135$

$$\text{Probability} = \frac{36 \times 30 + 9 \times 15}{9 \times 10^4} = \frac{4 \times 30 + 15}{10^4} = \frac{135}{10^4}.$$

Question 49. Let $(\lambda^2 + 1)x^2 - 4\lambda x + 2 = 0$ be a quadratic equation, then set of values of λ if exactly one root of quadratic equation lies in $(0, 1)$ is

(1) $(2, 3)$

(2) $(1, 3)$

(3) $[1, 2)$

(4) $(1, 3]$

Ans. (4)

Sol. $f(0).f(1) \leq 0$

$$\Rightarrow 2(\lambda^2 + 1 - 4\lambda + 2) \leq 0 \Rightarrow 2(\lambda^2 - 4\lambda + 3) \leq 0$$

$$(\lambda - 1)(\lambda - 3) \leq 0$$

$$\Rightarrow \lambda \in [1, 3]$$

But at $\lambda = 1$, both roots are 1 so $\lambda \neq 1$

Question 50. The orthocentre of $\triangle ABC$ where vertices are $A(-1, 7)$, $B(-7, 1)$, $C(5, -5)$ is

(1) $(-3, 3)$

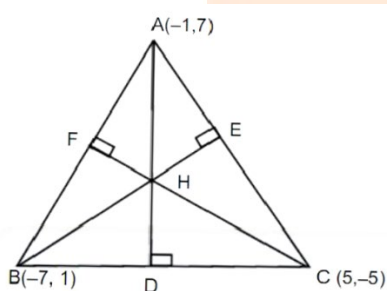
(2) $(3, -3)$

(3) $(3, 3)$

(4) $(-3, -3)$

Ans. (1)

Sol.



$$m_{BC} = \frac{6}{-12} = -\frac{1}{2}$$

∴ Equation of AD is $y - 7 = 2(x + 1)$

$$y = 2x + 9 \quad \dots(1)$$

$$m_{AC} = \frac{12}{-6} = -2$$

∴ Equation of BE is

$$y - 1 = \frac{1}{2}(x + 7)$$

$$y = \frac{x}{2} + \frac{9}{2} \quad \dots(2)$$

By (1) and (2)

$$2x + 9 = \frac{x + 9}{2}$$

$$\Rightarrow 4x + 18 = x + 9$$

$$\Rightarrow 3x = 9 \Rightarrow x = -3$$

$$\therefore y = 3$$

Question 51. m A.M. and 3 G.M. are inserted between 3 and 243 such that 2nd G.M. = 4th A.M. then $m =$.

Ans. (39)

Sol. 3, $A_1, A_2, A_3, \dots, A_m, 243$

$$d = \frac{243-3}{m+1} = \frac{240}{m+1}$$

3, G1, G2, G3, 243

$$r = \left(\frac{243}{3} \right)^{\frac{1}{3+1}} = (81)^{1/4} = 3$$

$$G2 = A4$$

$$\Rightarrow 3(3)^2 = 3 + 4 \left(\frac{240}{m+1} \right) \Rightarrow 27 = 3 + \frac{960}{m+1} \Rightarrow m+1 = 40$$

$$\Rightarrow m = 39$$

Question 52. A normal is drawn to parabola $y^2 = 4x$ at $(1, 2)$ and tangent is drawn to $y = e^x$ at (c, e^c) . If tangent and normal intersect at x-axis then find C.

Ans. 04.00

Sol. For $(1, 2)$ of $y^2 = 4x \Rightarrow t = 1, a = 1$

$$\text{Normal} \Rightarrow tx + y = 2at + at^3$$

$$\Rightarrow x + y = 3 \text{ intersect x-axis at } (3, 0)$$

$$y = e^x \Rightarrow \frac{dy}{dx} = e^x$$

$$\text{Tangent} \Rightarrow y - e^c = e^c (x - c)$$

At $(3, 0)$

$$\Rightarrow 0 - e^c = e^c (3 - c) \Rightarrow c = 4$$

Question 53. If relation $R_1 = \{(a, b) : a, b \in \mathbb{R}, a^2 + b^2 \in \mathbb{Q}\}$
and $R_2 = \{(a, b) : a, b \in \mathbb{R}, a^2 + b^2 \notin \mathbb{Q}\}$

Then

- (1) R_1 is transitive, R_2 is not transitive
- (2) R_1 is not transitive, R_2 is not transitive
- (3) R_1 is transitive, R_2 is transitive
- (4) R_1 is not transitive, R_2 is transitive

Ans. (2)

Sol. For R_1 , let $a = 1 + \sqrt{2}$, $b = 1 - \sqrt{2}$, $c = 8^{1/4}$

$$aR_1b \Rightarrow a^2 + b^2 = (1 + \sqrt{2})^2 + (1 - \sqrt{2})^2 = 6 \in \mathbb{Q}$$

$$bR_1c \Rightarrow b^2 + c^2 = (1 - \sqrt{2})^2 + (8^{1/4})^2 = 3 \in \mathbb{Q}$$

$$aR_1c \Rightarrow a^2 + c^2 = (1 + \sqrt{2})^2 + (8^{1/4})^2 = 3 + 4\sqrt{2} \notin \mathbb{Q}$$

$\therefore R_1$ is not transitive.

For R_2 , let $a = 1 + \sqrt{2}$, $b = \sqrt{2}$, $c = 1 - \sqrt{2}$

$$aR_2b \Rightarrow a^2 + b^2 = (1 + \sqrt{2})^2 + (\sqrt{2})^2 = 5 + 2\sqrt{2} \notin \mathbb{Q}$$

$$bR_2c \Rightarrow b^2 + c^2 = (\sqrt{2})^2 + (1 - \sqrt{2})^2 = 5 - 2\sqrt{2} \notin \mathbb{Q}$$

$$aR_2c \Rightarrow a^2 + c^2 = (1 + \sqrt{2})^2 + (1 - \sqrt{2})^2 = 6 \in \mathbb{Q}$$

$\therefore R_2$ is not transitive

Question 54. If the sum of first n terms of series

$$20 + 19\frac{3}{5} + 19\frac{1}{5} + 18\frac{4}{5} + \dots \text{is } 488 \text{ and } n\text{th term is negative}$$

then find n

(1) -4

(2) 4

(3) 1

(4) 6

Ans. (1)

Sol. Given series is: $\frac{100}{5} + \frac{98}{5} + \frac{96}{5} + \frac{94}{5} + \dots$

This forms an AP with $a = \frac{100}{5}$, $d = -\frac{2}{5}$

$$T_n < 0$$

$$\Rightarrow \frac{100}{5} + (n-1)\left(-\frac{2}{5}\right) < 0$$

$$\Rightarrow n > 51$$

$$488 = \frac{n}{2} \left[2\left(\frac{100}{5}\right) + (n-1)\left(-\frac{2}{5}\right) \right]$$

$$\Rightarrow n^2 - 101n + 2440 = 0$$

$$\Rightarrow n = 61 \text{ or } 40$$

Since $n > 51$, therefore $n = 61$

$$T_n = \frac{100}{5} + (61-1)\left(-\frac{2}{5}\right) = -4$$

Question 55. Surface area of cube is increasing at rate of 3.6 cm²/s. Find the rate at which its volume increases when length of side a is 10 cm.

- (1) 9
- (2) 10
- (3) 18
- (4) 20

Ans. (1)

Sol.

$$S = 6a^2 \Rightarrow \frac{dS}{dt} = 12a \cdot \frac{da}{dt} = 3.6 \Rightarrow 12(10) \frac{da}{dt} = 3.6 \Rightarrow \frac{da}{dt} = 0.03$$

$$V = a^3 \Rightarrow \frac{dV}{dt} = 3(10)^2 \cdot \left(\frac{3}{100}\right) = 9 \text{ cm}^3/\text{s}$$

Question 56. Which of the following point lies on plane containing lines

$$\vec{r} = \hat{i} + \lambda(\hat{i} + \hat{j} + \hat{k}) \text{ and } \vec{r} = -\hat{j} + \mu(-\hat{i} - 2\hat{j} + \hat{k}).$$

- (1) (1, 3, 6)

(2) (1, -3, 6)

(3) (-2, 1, 2)

(4) (1, 3, 1)

Ans. (2)

Sol. Normal of plane = $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ -1 & -2 & 1 \end{vmatrix}$

$\vec{n} = 3\hat{i} - 2\hat{j} - \hat{k}$

D.R. 's = 3, -2, -1

Plane $\Rightarrow 3(x - 1) - 2(y - 0) - 1(z - 0) = 0$

$\Rightarrow 3x - 2y - z - 3 = 0$

Question 57. $\lim_{x \rightarrow a} \frac{(a^2 + 2x^2)^{1/3} - (3x^2)^{1/3}}{(3a^2 + x^2)^{1/3} - (4x^2)^{1/3}} =$

(1) $\left(\frac{4}{3}\right)^{2/3}$

(2) $\frac{1}{3} \left(\frac{3}{4}\right)^{2/3}$

(3) $\frac{1}{3} \left(\frac{2}{3}\right)^{2/3}$

$$(4) \frac{1}{3} \left(\frac{4}{3} \right)^{2/3}$$

Ans. (4)

$$\begin{aligned} \text{Sol. } \lim_{x \rightarrow a} & \left(\frac{\frac{1}{3}(a^2 + 2x^2)^{-2/3} \cdot 4x - \frac{1}{3} \cdot (3x^2)^{-2/3} \cdot 6x}{\frac{1}{3}(3a^2 + x^2)^{-2/3} \cdot 2x - \frac{1}{3}(4x^2)^{2/3} \cdot 8x} \right) \\ &= \frac{\frac{1}{3}(3a^2)^{-2/3} \cdot a(4-6)}{\frac{1}{3}(4a^2)^{-2/3} \cdot a(2-8)} = \frac{3^{-\frac{2}{3}}}{4^{-\frac{2}{3}}} \cdot \frac{1}{3} \\ &= \frac{4^{\frac{2}{3}}}{3^{\frac{2}{3}}} \cdot \frac{1}{3} = \left(\frac{4}{3} \right)^{\frac{2}{3}} \cdot \frac{1}{3} \end{aligned}$$

Question 58. If $x^3 dy + xy dx = 2y dx + x^2 dy$ and $y(2) = e$ then $y(4) = ?$

$$(1) \frac{1}{2} + \sqrt{e}$$

$$(2) \frac{1}{2} \sqrt{e}$$

$$(3) \sqrt{e}$$

$$(4) \frac{3}{2} \sqrt{e}$$

Ans. (4)

Sol. $x^3 dy + xy dx + 2y dx + x^2 dy$

$$\Rightarrow (x^3 - x^2) dy = (2 - x) y dx$$

$$\Rightarrow \frac{dy}{y} = \frac{2 - x}{x^2 (x - 1)} dx$$

$$\Rightarrow \int \frac{dy}{y} = \int \frac{2 - x}{x^2 (x - 1)} dx \dots (i)$$

Let $\frac{2 - x}{x^2 (x - 1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x - 1}$

$$\Rightarrow 2 - x = Ax(x - 1) + B(x - 1) + Cx^2$$

$$\Rightarrow C = 1, B = -2 \text{ and } A = -1$$

$$\Rightarrow \int \frac{dy}{y} = \int \left\{ \frac{-1}{x} - \frac{2}{x^2} + \frac{1}{x - 1} \right\} dx$$

$$\Rightarrow \ln y = -\ln x + \frac{2}{x} + \ln |x - 1| + C$$

$$\therefore y(2) = e$$

$$\Rightarrow 1 = -\ln 2 + 1 + 0 + C$$

$$\Rightarrow C = \ln 2$$

$$\ln y = -\ln x + \frac{2}{x} + \ln |x - 1| + \ln 2$$

at $x = 4$

$$\Rightarrow \ln y(4) = -\ln 4 + \frac{1}{2} + \ln 3 + \ln 2$$

$$\Rightarrow \ln y(4) = \ln\left(\frac{3}{2}\right) + \frac{1}{2} = \ln\left(\frac{3}{2}e^{1/2}\right)$$

$$\Rightarrow y(4) = \frac{3}{2}e^{1/2}$$

Question 59. Find the number of 3 digit numbers if sum of their digits is 10

Ans. (55.00)

Sol. Let xyz be the three digit number

$$x + y + z = 10, x \leq 1, y \geq 0, z \geq 0$$

$$x - 1 = t \Rightarrow x = 1 + t \quad x - 1 \geq 0$$

$$t \geq 0$$

$$t + y + z = 10 - 1$$

$$t + y + z = 9, 0 \leq t, z, z \leq 9$$

$$\therefore \text{Non-negative integral solutions} = {}^{9+3-1}C_{3-1}$$

$$= {}^{11}C_2 = \frac{11 \cdot 10}{2} = 55$$

Question 60. If $\frac{a}{\cos \theta} = \frac{b}{\cos\left(\theta + \frac{2\pi}{3}\right)} = \frac{c}{\cos\left(\theta + \frac{4\pi}{3}\right)}$ then

find angle between vectors $a\hat{i} + b\hat{j} + c\hat{k}$ and $b\hat{i} + c\hat{j} + a\hat{k}$ if

$\theta = \frac{2\pi}{9}$ and $a^2 + b^2 + c^2 = 1$, is

- (1) $\frac{\pi}{3}$
- (2) $\frac{\pi}{6}$
- (3) $\frac{2\pi}{3}$
- (4) $\frac{5\pi}{6}$

Ans. (3)

Sol.

$$\begin{aligned} \frac{a}{\cos \theta} &= \frac{b}{\cos\left(\theta + \frac{2\pi}{3}\right)} = \frac{c}{\cos\left(\theta + \frac{4\pi}{3}\right)} \\ &= \frac{a + b + c}{\cos \theta + \cos\left(\theta + \frac{2\pi}{3}\right) + \cos\left(\theta + \frac{4\pi}{3}\right)} = \frac{a + b + c}{0} \end{aligned}$$

$$\Rightarrow a + b + c = 0 \Rightarrow a^2 + b^2 + c^2 + 2(ab + bc + ca) = 0$$

$$\Rightarrow ab + bc + ca = -\frac{1}{2}$$

Now let angle between given vectors is ϕ

$$\therefore \cos \phi = \frac{(a\hat{i} + b\hat{j} + c\hat{k}) \cdot (b\hat{i} + c\hat{j} + a\hat{k})}{a^2 + b^2 + c^2}$$

$$\cos \phi = \frac{ab + bc + ca}{1} = \frac{-1}{2}$$

$$\phi = \frac{2\pi}{3}$$

Question 61. If $(p \wedge q) \rightarrow (\sim q \vee r)$ has truth value false then the truth values of p, q, r respectively are

(1) T, T, F

(2) T, F, T

(3) F, F, T

(4) T, T, T

Ans. (1)

Sol. $(p \wedge q)$ should be TRUE and $(\sim q \vee r)$ should be FALSE.